Jiayi (Johnny) Li

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EDUCATION

University of Toronto 09/2026 - 06/2029 (Expected)
BS in Computer Science & Quantitative Biology Toronto, ON

- GPA: 3.78/4.0
- Relevant Coursework: Calculus with Proofs, Foundations of Comp Sci I & II, Writing Reports, Statistical Reasoning
- Activities: Biotechnology Innovation and Development, Science Communications, UofT Tri-Campus Hockey

RESEARCH EXPERIENCE

Radiation Dosimetry and Quality Assurance (QA) Research 01/2026 - Present
Undergraduate Researcher, Sunnybrook Hospital, under Dr. Arman Sarfehnia Toronto, ON

- Conducted research on radiation dosimetry and quality assurance to improve the accuracy and reliability of clinical dose measurements in radiotherapy
- Assisted in the design, testing, and evaluation of calorimetry-based detectors, contributing to improved detector robustness and reduced measurement error
- Supported the development of automated QA data analysis tools and performed literature reviews on dosimetry principles and clinical QA methodologies

Kidney Genetics and PAX2-Mediated Repair Research 11/2025 - Present
Undergraduate Researcher, Toronto General Hospital, under Dr. Moumita Barua Toronto, ON

- Independently reanalyzed podocyte mass spectrometry data to validate P3H2-mediated type IV collagen modifications, cross-referencing findings against 2 prior analyses to assess proteomic reproducibility
- Assisted in undifferentiated-to-differentiated podocyte cell culture and mass spectrometry workflows across experimental plates, gaining hands-on experience in protein extraction and sample preparation
- Performed Western blot to validate Col4a3 antibody specificity on podocyte protein extracts, supporting quality control for a key reagent in basement membrane integrity research

Cytomegalovirus (CMV) and Environmental Metal Exposure Research 03/2025 - 04/2026
Student Researcher, Northwestern University, under Dr. Hui Zhang Remote, ON

- Analyzed 4 heavy metal biomarkers across ≈550 U.S. adults from NHANES 2017–2018 to quantify associations between environmental metal exposure and persistent CMV infection, a virus affecting >50% of the U.S. population
- Applied and built 5+ statistical models in R including multivariable regression, GAMs, and sensitivity analyses across all metal-CMV exposure-outcome pairs to control for confounders
- Sole-authored research paper peer-reviewed and published by Asia Pacific Science Press in the Advances in Management and Intelligent Technologies (AMIT) journal

Parkinson's Disease (PD) and Protein Aggregation Research 07/2024 - 02/2026
Student Researcher, Harvard Medical School, under Dr. Ulf Dettmer Boston, MA

- Performed 3 IncuCyte live-cell assays at Brigham & Women's Hospital, quantifying α-synuclein aggregation across 5 compounds, 3 doses, and 35+ replicates per condition
- Conducted quantitative image analysis of inclusion burden, YFP signal, and cell density across 500+ data points to identify small molecules that disaggregate lipid-rich αS inclusions
- First-author peer-reviewed publication in the Journal of High School Science; selected for poster presentation at the 2026 AAAS Annual Meeting; delivered an oral talk at the CCIR Student Symposium to University of Cambridge faculty and Nobel Laureate Thomas R. Cech

PUBLICATIONS

- Li, J & Dettmer, U. (2025). Parkinson's Disease-linked protein α-synuclein: do small-molecule inhibitors of protein aggregation also prevent its accumulation with lipids?. *Journal of High School Science*, *9*(1). Retrieved from https://doi.org/10.64336/001c.131827
- Li, J. (2026). Cytomegalovirus infection in population samples: Are whole-blood levels of cadmium, mercury, selenium, and manganese associated with CMV serostatus? *Advances in Management and Intelligent Technologies*, *2*(1). https://doi.org/10.62177/amit.v2i1.1094
- Han, Z., Li, J., Yang, S., … Shi, J. (2026). *Towards a nonlinear future: How to maximize personal value and achieve true growth in the age of AI*. China Financial & Economic Publishing House.

WORK EXPERIENCE

NorthStar Special Needs Society (NSNS) 09/2025 - Present
Lead Instructor & Coach Toronto, Canada

- Coaching children aged 5–14 in skating and hockey fundamentals, adapting instruction with clear communication for diverse learning needs
- Leading multiple Learn to Skate and hockey camps for 20+ children, designing engaging drills and games to enhance skill development and enjoyment
- Supporting children with autism in developing confidence, coordination, and social skills through skating and hockey.

My Learning Space AI (MYLS) 06/2025 - Present
AI Product Manager Intern Toronto, Canada

- Conducting A/B testing and user interviews to identify key friction points and drive improvements in monetization
- Creating AI-generated mock prototypes to test concepts and support early-stage user research with target learners
- Designing and iterating an optimized booking flow, improving conversion and successful bookings by 9%.
- Collaborating cross-functionally with engineering and design to implement data-backed product improvements

Pharmaron Beijing Co., Ltd. 05/2025 - 08/2025
Chemical Analysis Researcher Beijing, China

- Operated Liquid Chromatography–Mass Spectrometry (LC-MS) to analyze over 50 pharmaceutical samples, processing and interpreting mass spectra using Excel for compound identification and purity assessment.
- Collaborated with 8 professional chemists to develop and synthesize compounds used in widely distributed medications, including cold treatments and acetaminophen (Tylenol).
- Prepared and delivered 10+ research presentations translating LC–MS drug composition data into insights for international clients.

PROJECTS

S&P 500 Market Crash Simulator – Python

- Led a 4-person team to build an interactive financial contagion simulator modeling crash propagation across all 503 S&P 500 companies using a weighted graph with ~24,000 Pearson correlation edges ($r \geq 0.7$) derived from COVID-19 period log-returns
- Engineered a recursive contagion algorithm in Python that attenuates crash impact across correlated neighbors, enabling real-time simulation of user-defined crash scenarios with live market cap loss estimates across affected companies
- Built a full-stack interactive web dashboard using Dash and Plotly, integrating yfinance historical data retrieval, NetworkX graph layout computation, and NumPy correlation analysis into a single deployable pipeline

Towards a Nonlinear Future – Book Publication (China Financial & Economic Publishing House)

- Co-authored a 5-chapter analytical volume exploring artificial intelligence's structural impact on education, skill formation, and institutional decision-making in North America
- Contributed research and writing on AI-education interactions, university-level adaptations, and future skill frameworks, emphasizing interdisciplinary cognition, systems thinking, and ethical responsibility
- Synthesized qualitative case studies and policy-level observations into an accessible framework connecting AI theory with practical educational and career planning outcomes

Nucleus Independent Living Services - Relating Chronic Conditions to Services

- Led a 3-person team in developing a 22-page statistical analysis evaluating how clients with varying chronic conditions respond to Nucleus services
- Analyzed >5,000 client records to quantify changes in hospital days, healthcare costs, and readmission rates using data wrangling, visualization, and summary statistics in R.

SKILLS AND INTERESTS

- **Technical Skills:** Python, HTML/CSS, Javascript, R, Microsoft Offices, CAD, Latex
- **Language Skills:** English, Mandarin (Fluent), Spanish (Intermediate), French (Intermediate)
- **Interests:** Leadership, Hockey, Robotics, Research, Violin, Chemistry, Card Games, Magic